



BLUESTOCK

MUTUAL FUND ANALYTICS

Capstone Project

SUBMITTED BY

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Python



Pandas



SQLite



SQL



Power BI



GitHub

June 2026

Project Overview



Problem Statement

Mutual fund datasets are spread across multiple sources and require cleaning, integration, analysis, and visualization to generate meaningful insights.



Project Goal

Build an end-to-end analytics pipeline that:

- Processes mutual fund datasets
- Stores data in SQLite
- Performs analytics
- Creates business insights
- Visualizes results using Power BI

From Data to Decisions



Process Datasets



Store in SQLite



Perform Analytics



Generate Insights



Visualize in Power BI

Dataset Overview

Dataset	Purpose
Fund Master	Scheme Information
NAV History	Historical NAV
AUM Data	Assets Under Management
SIP Inflows	SIP Trends
Category Inflows	Category Analysis
Folio Count	Investor Growth
Scheme Performance	Returns
Transactions	Investor Activity
Portfolio Holdings	Sector Analysis
Benchmark Indices	Benchmark Comparison



10

CSV Files

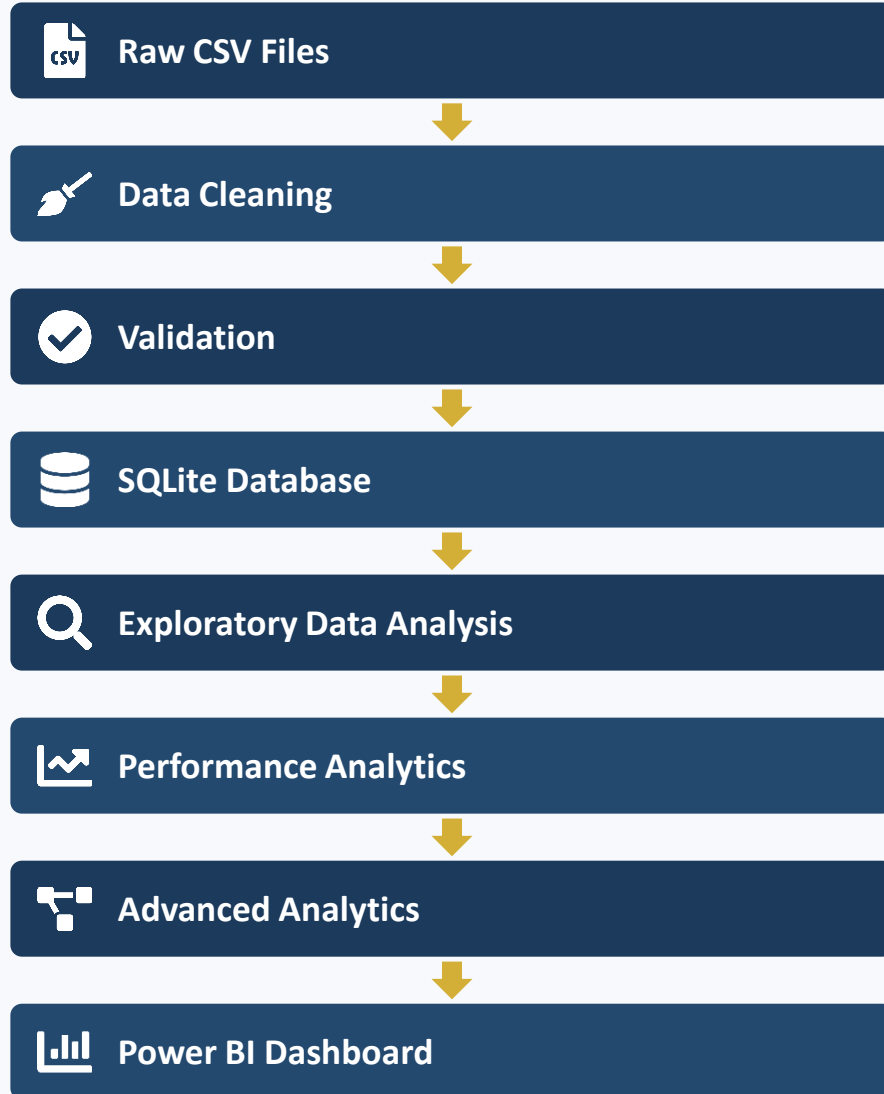
Database

SQLite Database

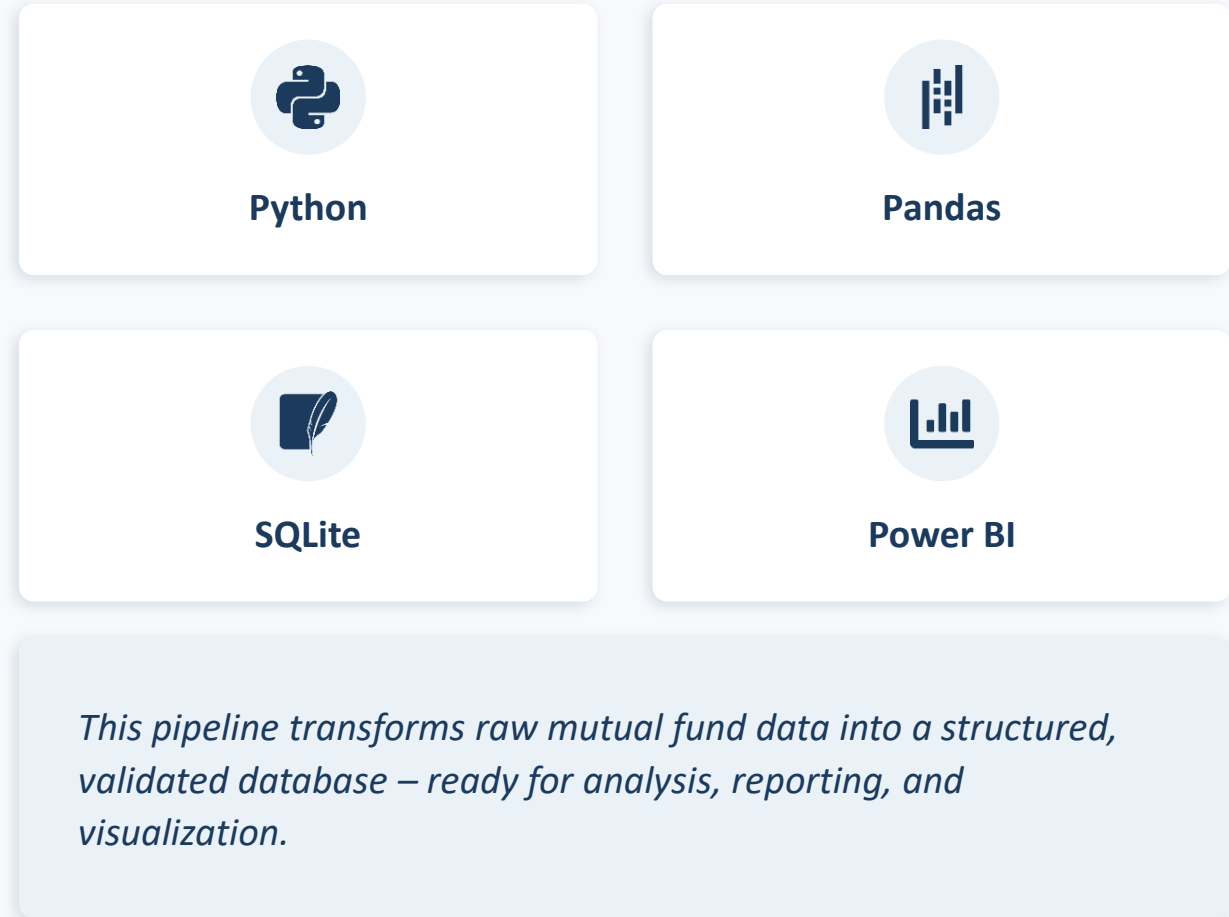
bluestock_mf.db

Total Datasets: 10 CSV Files

Project Architecture



Tools Used



Data Cleaning & ETL



Data Cleaning Tasks



Missing value treatment



Duplicate removal



Date standardization



Column normalization



Validation checks

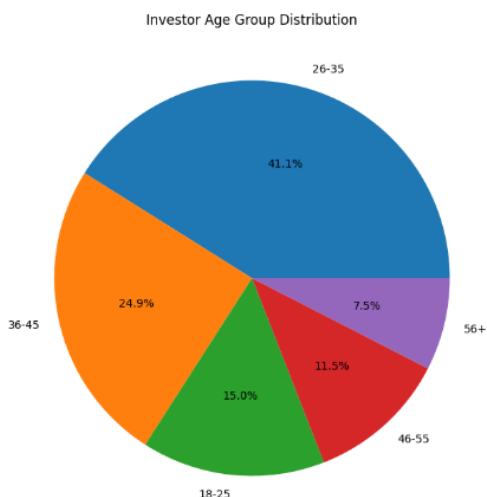


ETL Pipeline Scripts

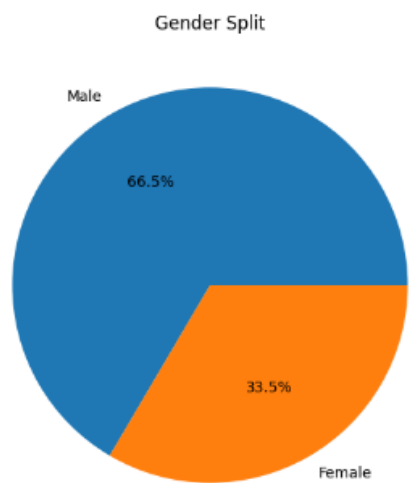
```
✓ scripts
  ├── amfi_validation.py
  ├── data_cleaning.py
  ├── date_dimension.py
  ├── etl_load.py
  └── etl_pipeline.py
```

Modular ETL scripts handle data cleaning, AMFI validation, date-dimension setup, and loading into the SQLite database.

Investor Demographics



Investor Age Group Distribution



Gender Split

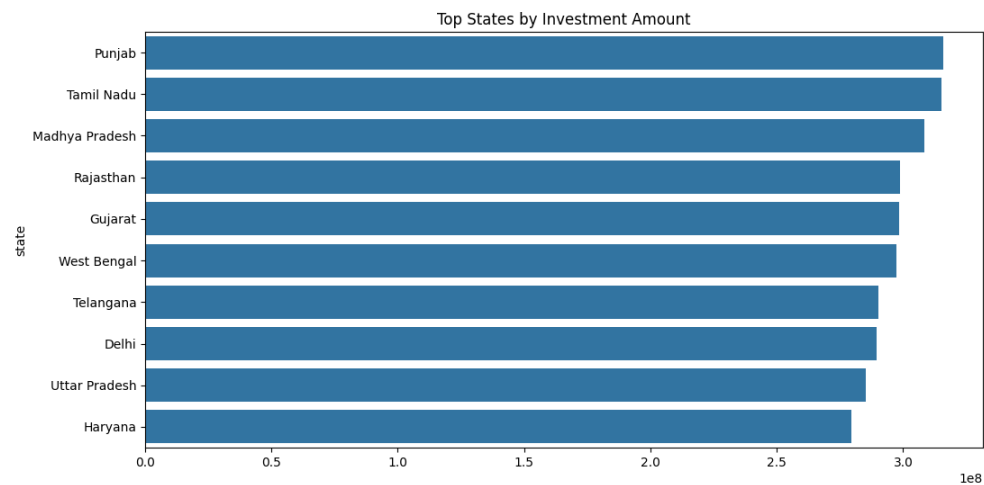


Key Findings

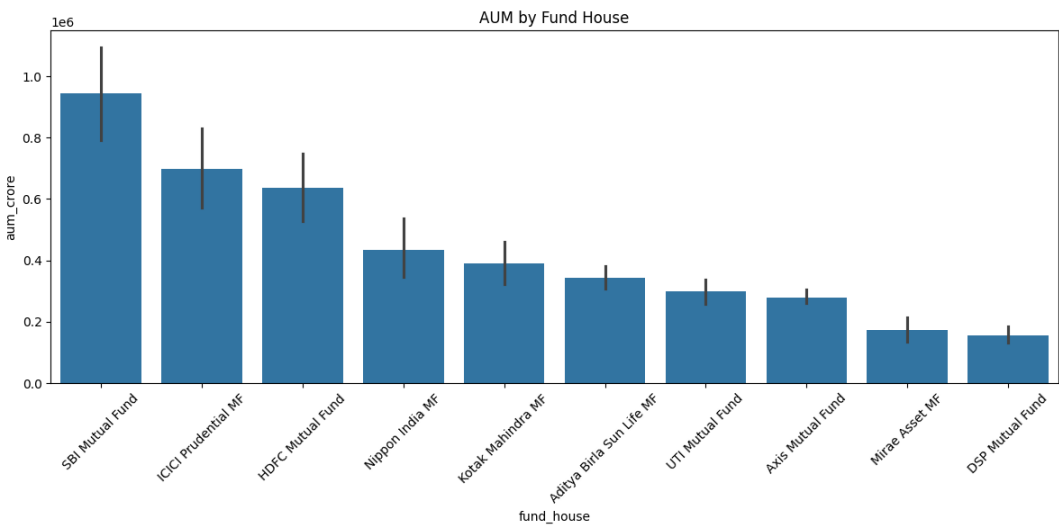
- Investors aged 26–35 form the largest group at 41% of the base.
- Working-age investors (18–45) account for over 80% of all participants.
- Male investors represent 66.5% vs. 33.5% female participation.

Demographic insights help tailor product design, marketing, and SIP outreach toward the most engaged investor segments.

AUM & Geographic Trends



Top States by Investment Amount



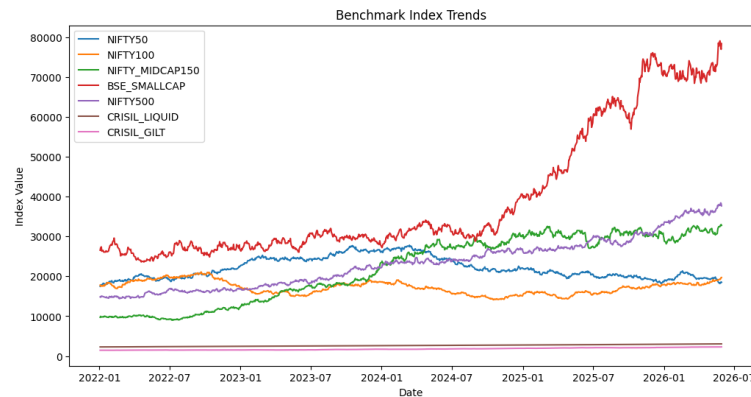
AUM by Fund House



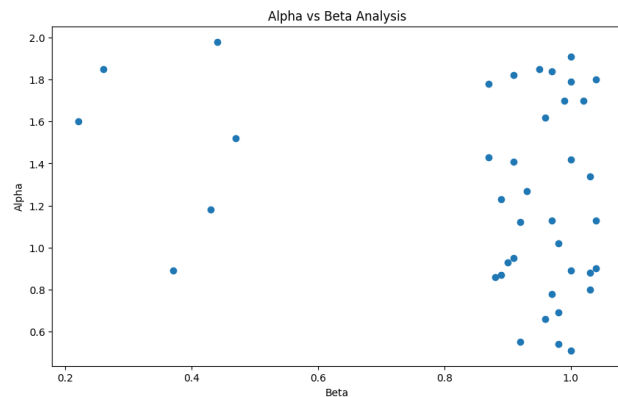
Key Findings

- Punjab, Tamil Nadu, and Madhya Pradesh lead all states in total investment amount.
- SBI Mutual Fund commands the highest average AUM, ahead of ICICI Prudential and HDFC.

Performance Analytics



Benchmark Index Trends



Alpha vs Beta Analysis

Metrics Calculated



CAGR



Alpha



Beta



Sharpe Ratio



Volatility



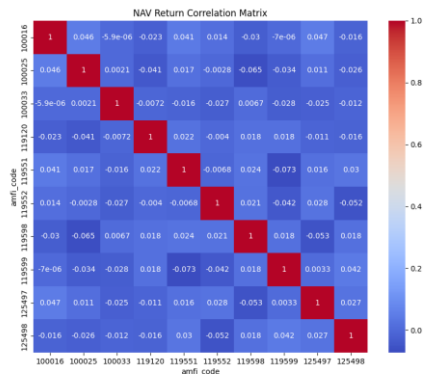
Maximum Drawdown



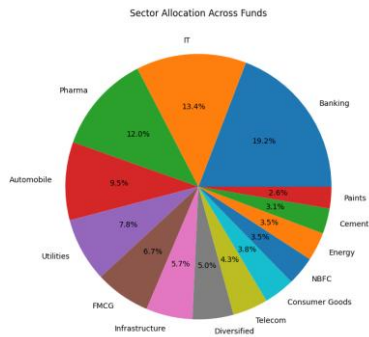
Insight

An average Alpha of 1.25 and Beta of 0.87 show funds delivering excess returns with below-market volatility – a strong risk-adjusted profile.

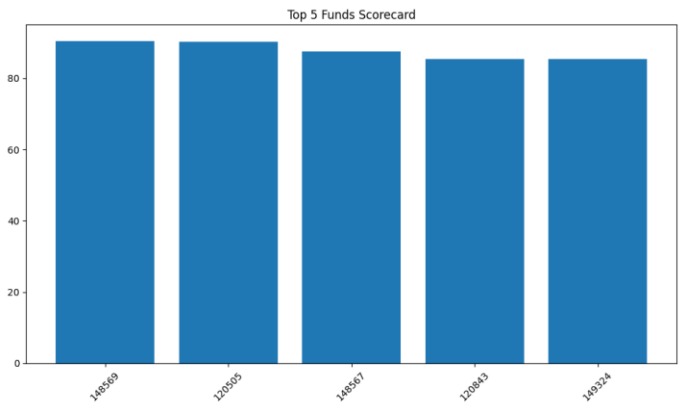
Advanced Analytics



NAV Return Correlation



Sector Allocation



Top 5 Funds Scorecard

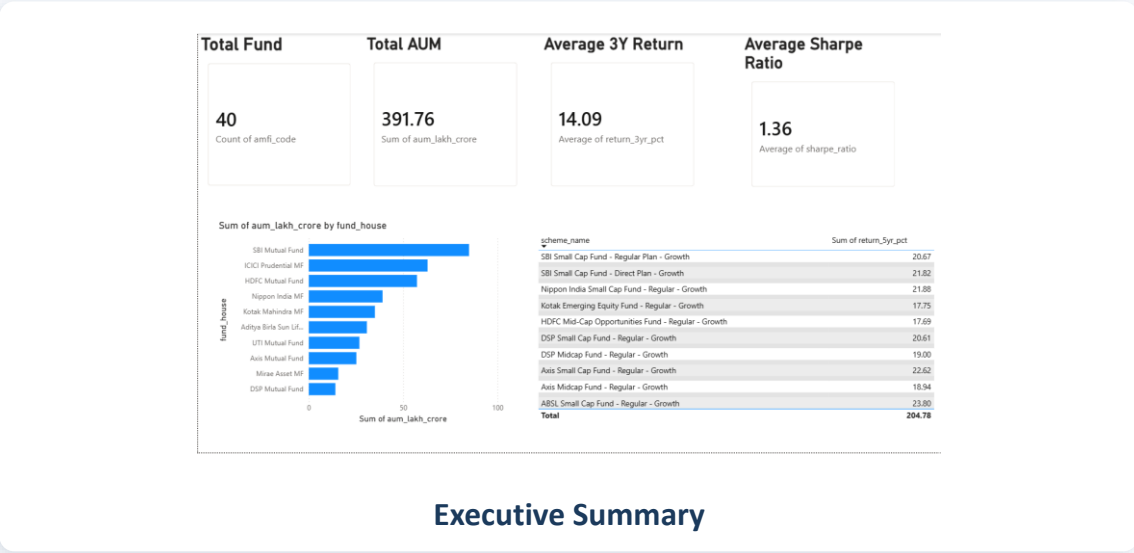
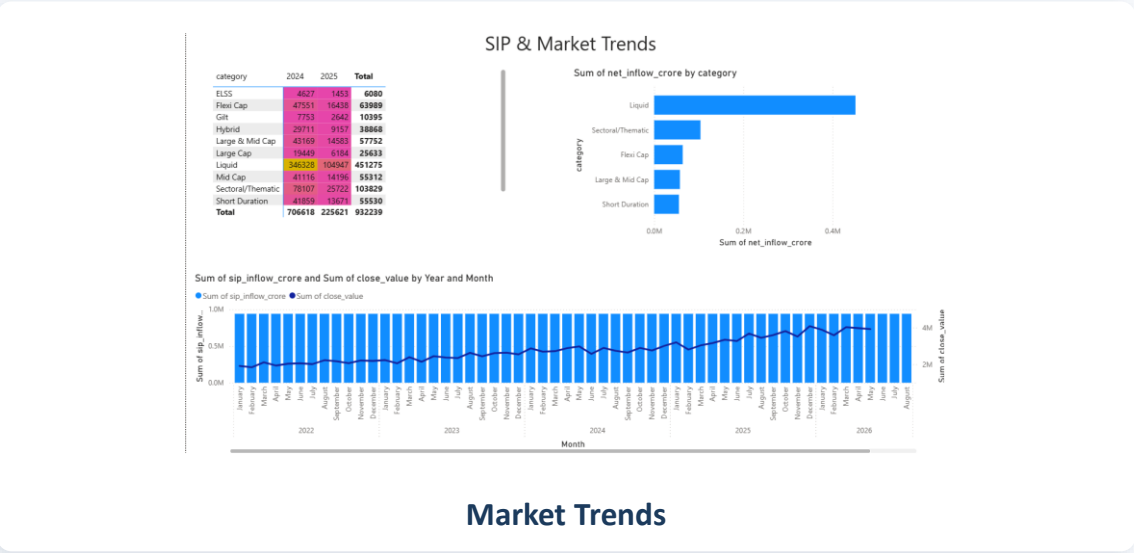
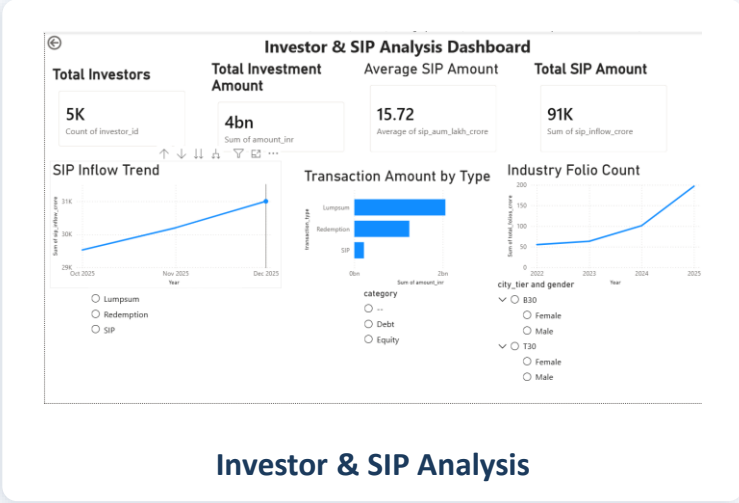
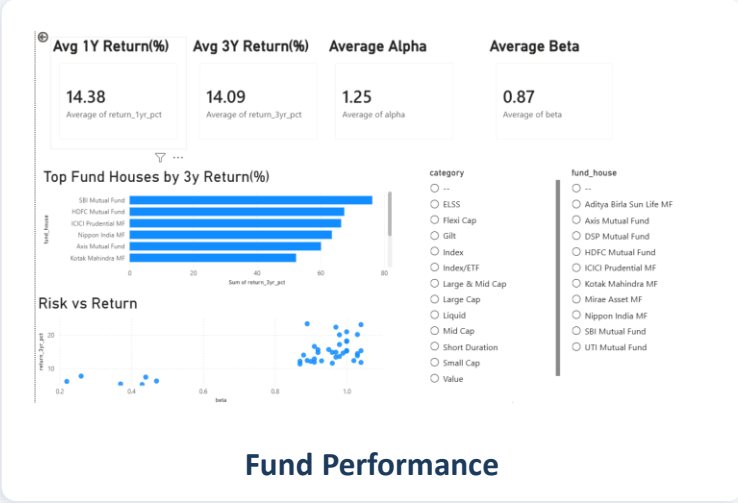
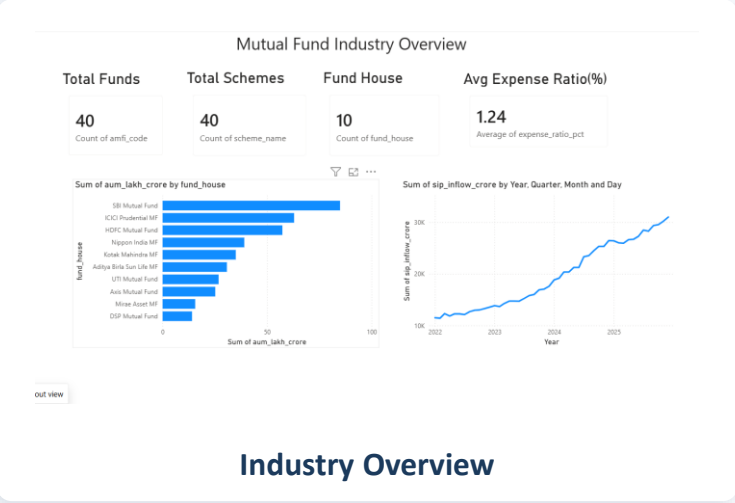
Analyses Performed

-  Correlation Analysis
-  Fund Ranking
-  Sector Allocation Analysis

Findings

- Equity-oriented funds showed stronger performance.
- Sector allocation highlights diversification patterns, led by Banking, IT, and Pharma.
- Low NAV correlations across top funds suggest healthy diversification.

Power BI Dashboard



Key Findings



AUM has grown consistently, reaching ₹391.76 lakh crore across 40 tracked schemes.



SIP participation continues to rise, with total inflows of ₹91K crore.



Equity funds dominate investor preference, driving sustained retail inflows.



Top funds outperform benchmarks, averaging 14.09% 3Y returns and a 1.36 Sharpe ratio.



Fund ranking highlights consistent leaders such as SBI and ICICI Prudential.



The dashboard enables data-driven decisions across 40 schemes from 10 fund houses.

Conclusion

PROJECT ACHIEVEMENTS

- ✓ ETL Pipeline Built
- ✓ EDA Completed
- ✓ Advanced Analytics Completed
- ✓ SQLite Database Created
- ✓ Performance Analytics Completed
- ✓ Power BI Dashboard Developed



Outcome

Successfully built an end-to-end mutual fund analytics platform using modern data analytics tools – from raw data to business-ready insights.



Thank You

Questions & Discussion

Harsh Gupta